

$$1) \text{ Attn params} = \underbrace{DNH + 2DKH + NHD}_{\text{attn proj}} + \underbrace{D}_{\text{layer norm}}$$

$$= DNH + 2DNH + DNH + D = 4DNH + D$$

$$\text{MLP params} = 2DF + FD = 3DF$$

$$\text{Total} = (4DNH + D + 3DF) \cdot L + 2DV$$

$$= (4 \cdot 5120 \cdot 40 \cdot 128 + 5120 + 3 \cdot 5120 \cdot 13824) \cdot 40 + 2 \cdot 5120 \cdot 32000$$

$$\approx 13 \times 10^9 = 13B \text{ params}$$

2) BS = 16M tokens, using Adam, store params in b516, opt state in fp32, checkpoint act 3 times/layer

$$\text{Param mem} = 2 \cdot 13 \times 10^9 \text{ bytes}$$

$$\text{Opt mem} = \underbrace{4 \cdot 13 \times 10^9 + 4 \cdot 13 \times 10^9}_{1st + 2nd \text{ moment}}$$

$$\text{Act ckpt mem} = 2L \cdot \left(\underbrace{2BF}_{\text{gating up proj}} + \underbrace{BD}_{\text{down proj}} \right) = 2B(2F + D)$$

$$= 2 \cdot 40 \cdot 16 \cdot 10^6 (2 \cdot 13824 + 5120)$$

$$\text{Total} = \text{Param} + \text{Opt} + \text{Act} \approx 4.2 \times 10^{13} \text{ bytes} = 42TB$$

3. Train with:

$$T = 32K$$

$$BS = 3M$$

on TPU v5p 16x16x16 slice (has wraparound counts)

bfloat16 weights, fp32 optimizer

i) From prev q, assuming Adam optimizer, the model weights + opt state is 130 GB > 96 GB which is the v5p's memory capacity.

ii) With zero-3 type FSDP, we have no duplicated data.

$$\text{Bytes}_{\text{Total}} = (10 \cdot 13 \cdot 10^9) + 2 \cdot 2B(2F+D)$$

$$= (10 \cdot 13 \cdot 10^9) + 2 \cdot 40 \cdot (3 \cdot 10^6) \cdot (2 \cdot 13824 + 5120)$$

$$\approx 7.99 \cdot 10^{12} \text{ bytes} \approx 8 \text{ TB}$$

Total v5p 16x16x16 slice has $16^3 \cdot 96 \text{ GB} = 393 \text{ TB}$, so we are in the hbm limit. Each device will have $(8 \text{ TB}) / 16^3 : 1.96 \text{ GB}$ used.
($x = 16^3$)

$$T_{\text{comm}} = \frac{2 \cancel{2BDF}}{3 \cdot w_{ii}}$$

$$T_{\text{math}} = \frac{2 \cancel{2BDF}}{x \cdot C}$$

$$\Rightarrow \frac{B}{x} = \frac{3 \cdot 10^6}{16^3} = 732 < \frac{4.59 \cdot 10^{14}}{3 \cdot 1.8 \cdot 10^{11}} = 850$$

Using pure FSDP, while it is possible, we

will be comm bounded so it's better to try FSDP + TP.

iii) a) To compute bound:

$$\frac{B}{N} > \frac{\kappa^2}{\underbrace{M_x M_y F}_{2 \text{ for 3D mesh}}} = \frac{(2550^2)}{2 \cdot 13824} \approx 235$$

$\frac{B}{N} = 850 > 235$, so we can use FSDP + TP.

b) Choosing M_x, M_y doesn't matter $\Rightarrow$ just changes X_{opt} .

Let $M_x = 2, M_y = 1$

$$X_{opt} = \sqrt{\frac{B}{F} \cdot \frac{M_x}{M_y} \cdot N} = \sqrt{\frac{(3 \cdot 10^6)}{13824} \cdot 2 \cdot 16^3} \approx 1333$$

$$\Rightarrow \text{mesh} = \{x: 1024, y: 4\}$$

need int + want wraparounds

FSDP + TP sharding:

$$In[B_x, D_y] \cdot_D W_{in}[D_x, F_y] \cdot_F W_{out}[F_y, D_x] \rightarrow Out[B_x, D_y]$$

Per device mem = 1.9 GB (same as pt b since there's no data duplication)

c) No redundant comp w/ FSDP + TP:

$$\text{FLOPS}_{\text{Total}} = 6 \cdot 133 \cdot 3M = 6 \cdot 13 \cdot 10^9 \cdot 3 \cdot 10^6$$

$$T_{\text{comp}} = \frac{(6 \cdot 13 \cdot 10^9 \cdot 3 \cdot 10^6)}{16^3 \cdot 4.59 \cdot 10^{14} \cdot 0.4} = 0.3$$